Done

Unesco Heritage Sites Many-to-One

In this assignment you will read some **Unesco Heritage Site (https://whc.unesco.org/en/list/)** data in comma-separated-values (CSV) format and produce properly normalized tables as specified below. Once you have placed the proper data in the tables, press the button below to check your answer.

Host: pg.pg4e.com

Port: 5432

Database: pg4e_3774c65c52

User: pg4e_3774c65c52

Password: (hide/show copy)

Check Answer

Here is the the **psql** command to access your database:

```
psql -h pg.pg4e.com -p 5432 -U pg4e_3774c65c52 pg4e_3774c65c52
```

Here is the structure of the tables you will need for this assignment:

```
DROP TABLE unesco_raw;
CREATE TABLE unesco_raw
 (name TEXT, description TEXT, justification TEXT, year INTEGER,
    longitude FLOAT, latitude FLOAT, area_hectares FLOAT,
    category TEXT, category_id INTEGER, state TEXT, state_id INTEGER,
    region TEXT, region_id INTEGER, iso TEXT, iso_id INTEGER);

CREATE TABLE category (
  id SERIAL,
  name VARCHAR(128) UNIQUE,
  PRIMARY KEY(id)
);

... More tables needed
```

To load the CSV data for this assignment (whc-sites-2018-small.csv?PHPSESSID=8588d29f27c2fca8596f6bf2d4cc6647") use the following **copy** command. Adding **HEADER** causes the CSV loader to skip the first line in the CSV file. The **\copy** command must be one long line.

```
\copy unesco_raw(name,description,justification,year,longitude,latitude,area_hectares,category,sta
Done
```

Normalize the data in the **unesco_raw** table by adding the entries to each of the lookup tables (category, etc.) and then adding the foreign key columns to the **unesco_raw** table. Then make a new table called **unesco** that removes all of the un-normalized redundant text columns like **category**.

If you run the program multiple times in testing or with different files, make sure to empty out the data before each run.

The autograder will look at the **unesco** table.

To grade this assignment, the program will run a query like this on your database and look for the data it expects to see:

```
SELECT unesco.name, year, category.name, state.name, region.name, iso.name
  FROM unesco
  JOIN category ON unesco.category_id = category.id
  JOIN iso ON unesco.iso_id = iso.id
  JOIN state ON unesco.state_id = state.id
  JOIN region ON unesco.region_id = region.id
  ORDER BY category.name, unesco.name
  LIMIT 3;
```

The expected result of this query on your database is:

Name	Year	Category	State	Region	iso
Khomani Cultural Landscape	2017	Cultural	South Africa	Africa	za
al Saflieni Hypogeum	1980	Cultural	Malta	Europe and North America	mt
ingvellir National Park	2004	Cultural	Iceland	Europe and North America	is